

Peatfr: An R package to forecast tropical peatland fire risk with stochastic, machine learning, and optimisation methods

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ABSTRACT

Early detection of tropical peatland fire is crucial to anticipate and mitigate fire risks effectively. However, existing software packages for predicting the occurrence of fires often lack comprehensive integration of methods and techniques, which potentially limits their application. Here, we propose *peatfr*, a novel R package to forecast tropical peat fire risk with stochastic, machine learning, and optimisation methods. The *peatfr* is designed with three main functions, which comprise of data imputation, time series forecasting, and fire risk prediction. The data imputation process addresses missing or incomplete time series data, ensuring that the dataset remains reliable for the subsequent analysis and forecasting. This package provides stochastic and machine learning methods, including ARIMA with Box-Cox transformation, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), to forecast time series data that significantly influence tropical peatland fire. The fire risk is calculated through Peat Fire Vulnerability Index, which employs Nelder-Mead optimisation method to obtain the optimal parameters. This approach allows *peatfr* to update the parameters automatically based on the input data, which results in a reliable package. Furthermore, because the *peatfr* operates independently without relying on external software to forecast the fire event, this package becomes straightforward and user-friendly. The capability to update parameters automatically and self-contained framework enhance its potential as an early warning tool for tropical peatland fire. We demonstrate the application of *peatfr* to predict the occurrence of tropical peatland fire in Sabangau, Central Kalimantan, Indonesia.

1. Introduction

Tropical peatland fires have serious impact on the environment, public health, and economic stability. Early detection and prediction of peat fires will benefit to anticipate potential hazards and to implement an effective risk management. The purpose of this paper is to develop a novel R package, *peatfr*, for predicting fire risk in the tropical peatland with stochastic, machine learning, and optimisation methods. The *peatfr* can update the parameters automatically based on the input data and does not depend on the external software, which results in a more comprehensive tool for early detection of tropical peatland fire. Fire danger in *peatfr* is represented by the Peat Fire Vulnerability Index (PFVI), a metric developed and validated for predicting fire occurrence in tropical peatlands. The PFVI integrates water table, soil moisture, rainfall, and air temperature into a unified indicator of peat fire susceptibility. The package is openly available through the GitHub

repository at <https://github.com/mellygsl/peatfr>.

Tropical peatlands have a critical role in the global carbon cycle (Leng et al., 2018; S. Page et al., 2022; Ribeiro et al., 2021) and provide vital ecosystem services to the environment (Mishra et al., 2021). High rates of peat production from organic matter supported by saturated conditions due to low hydraulic gradient (Baird et al., 2017; Kurnianto et al., 2015) result in a carbon-rich ecosystem. Tropical peatlands protect the environment from floods and drought because peat can maintain and regulate the water balance (Lupascu et al., 2020). Moreover, the unique characteristics of the tropical peatlands provide a suitable habitat for rare species, for example, the orangutan and Sumatran tiger (Girkin et al., 2022).

The condition changes dramatically for damaged tropical peatlands due to fire. Tropical peatland fires produce massive environmental and human health issues, which lead to significant economic losses. For example, the intense haze caused by peatland fires killed many people

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across Indonesia, Malaysia, Singapore, and Brunei in 2015 (Koplitiz et al., 2016; Marlier et al., 2015). This phenomenon released about 0.81–2.57 Gt C into the atmosphere, which is equivalent to around 40 % of total global fossil fuel emissions in that year (Huijnen et al., 2016; Lohberger et al., 2018). The estimated economic loss for Indonesia due to this peatland fire is about US\$ 16 billion, considering the destruction of infrastructure, property, and agriculture (Horton et al., 2021).

An early warning system is essential in addressing tropical peatland fires due to its ability to anticipate and mitigate fire risks effectively. Early detection enables stakeholders to implement targeted measures, including, deploying firefighting resources, constructing firebreaks, or issuing public advisories, which in turn minimise the intensity and extent of fires. Furthermore, by preventing extensive fires, an early warning system reduces the financial burden associated with healthcare costs due to haze related illnesses, and economic losses from disrupted agriculture or infrastructure (Kiely et al., 2021; Situngkir et al., 2025). The effectiveness of an early warning system for tropical peatland fires heavily depends on the integration of software capable of predicting fire risks accurately and efficiently. Such software acts as the analytical backbone of the system, which determines fire prone conditions and provides actionable insights. The need for predictive software arises from the dynamic and complex nature of tropical peatlands fire, which involves the interactions and feedback from climate, water table, and fuel (Cobb and Harvey, 2019; Leng et al., 2018; S. E. Page and Hooijer, 2016; Taufik et al., 2019).

Although an early warning system is crucial for preventing the impact of tropical peatland fire, there is a lack of comprehensive software to support this system. For example, Widyastuti et al. (2021) developed PeatFire software to analyse the interaction between fire

ignition, spreading, and peat characteristics in tropical peatland through agent-based model. The simulations result from this software indicated that the water table depth, peat dryness index, and number of dry days before ignition provides significant influence on the scale of burnt area. However, PeatFire software does not have the ability to update the parameters automatically based on the input data, which limits its application as early warning tools for fire forecasting. Mezbahuddin et al. (2023) combined ecohydrological software (M. Mezbahuddin et al., 2014, 2015) with artificial neural network in predicting tropical peatland fire. The inclusion of hydrological variables, including, water table depth and soil moisture, improve the accuracy of the software to estimate the occurrence of fires. However, this software depends on external weather or seasonal forecasts to function as an early warning tool for tropical peatland fire.

This paper introduces a new R package *peatfr*, which was developed to address these limitations, by providing a comprehensive and user-friendly software to predict the tropical peatland fire through stochastic, machine learning, and optimisation methods. Package *peatfr* uses data that significantly influences peatland behaviour, including, water table, soil moisture, rainfall, and air temperature (Mahdiyasa et al., 2022, 2023; S. Mezbahuddin et al., 2023; Widyastuti et al., 2021; Young et al., 2023), obtained from direct measurements from the field or indirect measurements through the satellite. Despite some of the functions discussed are available across various statistical software and packages (Hyndman and Khandakar, 2008; Templ et al., 2011, 2012), *peatfr* integrates data imputation, time series forecasting through stochastic and machine learning methods, and optimisation functions with the fire risk index in the tropical peatland, providing the essential tools to describe and predict the occurrence of tropical peatland fire, which are not

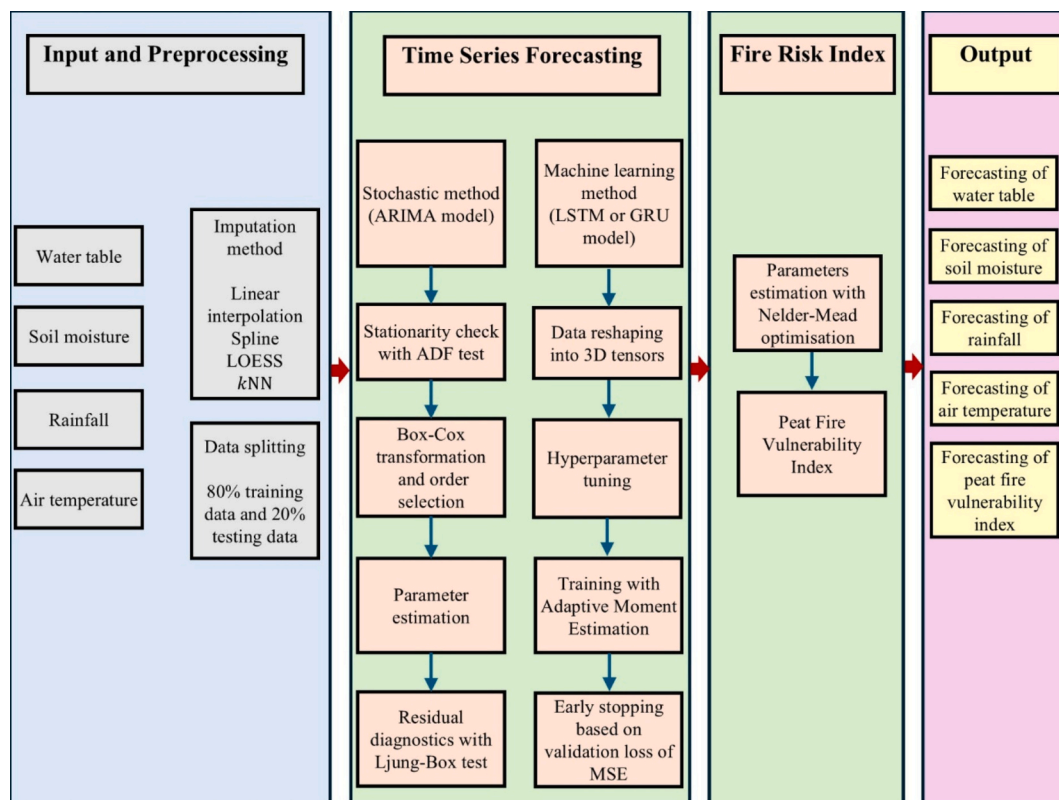


Fig. 1. The *peatfr* package framework consisting of input and preprocessing, time series forecasting, fire risk index, and output. This package requires time series data of water table, soil moisture, rainfall, and air temperature. The data imputation process employs four methods, including, linear interpolation, spline, locally estimated scatterplot smoothing (LOESS), and *k*-nearest neighbour (kNN). The time series forecasting uses ARIMA stochastic model and Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) machine learning models. The calculation of fire risk index is conducted through Peat Fire Vulnerability Index (PFVI) and Nelder-Mead optimisation. The package produces the forecasting of water table, soil moisture, rainfall, air temperature, and PFVI, which is crucial for early warning system of tropical peatland fire.

readily available in other software sources.

2. Methods

The *peatfr* package is structured into four main functional components (Fig. 1), each designed to address specific aspects of data preparation, forecasting, and fire risk assessment in tropical peatlands. First, the input and preprocessing component manages data preparation through two essential procedures, which include data imputation and data splitting. The data imputation module is designed to address missing or incomplete time series, ensuring the continuity and consistency of hydrological and meteorological records prior to model training. Following data refinement, the dataset is systematically divided into 80 % training and 20 % testing subsets to enable model calibration, performance evaluation, and assessment of generalisation capability. Second, the time series forecasting component performs predictive modelling through both stochastic and machine learning approaches. The stochastic module employs the ARIMA model, while the machine learning module utilises Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks. Third, the fire risk index component estimates the Peat Fire Vulnerability Index (PFVI), whose parameters are optimised using the Nelder–Mead algorithm to enhance the accuracy and robustness of fire risk estimation. This approach enables the PFVI to dynamically capture interactions among variables, including water table, soil moisture, rainfall, and air temperature. Finally, the output component generates forecasts for hydrological and meteorological variables and the PFVI. These outputs provide valuable insights for early warning systems, fire danger mapping, and decision support in peatland management and restoration planning.

2.1. Data preparation in *peatfr*

Accurate data preparation is a crucial step in ensuring the reliability of time series modelling and fire risk prediction. In the context of tropical peatland monitoring, hydrological and meteorological datasets often contain missing or inconsistent observations due to instrument failure, transmission errors, or harsh field conditions (Colston et al., 2018; Ma et al., 2024). To address these issues, the *peatfr* package includes a comprehensive data preprocessing framework that enhances data continuity and quality prior to model training and forecasting. An essential component of this framework is the data imputation process, which reconstructs missing values using statistically and computationally efficient techniques. The *peatfr* provides four methods for data imputation process, including, linear interpolation, spline, locally estimated scatterplot smoothing (LOESS), and *k*-nearest neighbour (*k*NN) (Lin and Tsai, 2020; Ren et al., 2023; Sun et al., 2023) as summarised in Table 1.

Linear interpolation is a straightforward method for imputing missing data by connecting two adjacent known values with a straight line and estimating the intermediate points accordingly. It preserves local continuity and is especially suited for time series data with relatively small gaps. Its main advantage is simplicity, requiring minimal computational effort and assumptions. However, it captures neither complex nonlinearities nor correlations with other variables, potentially leading to inaccuracies if the data exhibits large fluctuations or more intricate patterns.

Spline imputation in the *peatfr* package employs cubic polynomial functions to approximate missing data, which ensures continuity and smoothness across the boundaries between segments. The cubic function enables the imputation process to capture nonlinear patterns and subtle curvatures in the data, which may be overlooked by simpler linear approaches. The placement and number of knots play a pivotal role in controlling the smoothness of the estimated function, which in turn influencing imputation accuracy. However, splines method might introduce unwanted oscillations or overshooting in segments with sparse or noisy observations.

Table 1

Summary of the advantages, limitations, and rationale for selecting the imputation methods, including linear interpolation, spline, locally estimated scatterplot smoothing (LOESS), and *k*-nearest neighbour (*k*NN).

Method	Advantages	Disadvantages	Rationale for selection
Linear interpolation	Simple, fast, and easy to implement; preserves local continuity between adjacent data points.	Assumes linear transitions and cannot capture nonlinear or abrupt variations.	Suitable for variables exhibiting gradual and near-linear temporal changes.
Spline	Captures nonlinear patterns and ensures smooth transitions with continuous first and second derivatives; adaptable to various data shapes.	May introduce oscillations near knots and requires careful knot placement as well as higher computational effort.	Appropriate for continuous data showing seasonal or curvilinear trends.
LOESS	Highly flexible and capable of modelling complex, nonlinear relationships through locally weighted regression.	Computationally intensive, sensitive to bandwidth selection, and prone to overfitting or boundary effects.	Effective for capturing dynamic and nonlinear temporal variations.
<i>k</i> NN	Utilises multivariate relationships, is nonparametric and distribution-free, and adaptable to mixed-type data.	Computationally demanding for large datasets and sensitive to the choice of <i>k</i> and distance metric.	Suitable for capturing and utilising correlations across multiple variables.

The *peatfr* package provides LOESS method that can be employed for data imputation by fitting a smooth curve to subsets of the data around each missing observation. This localised fitting allows LOESS to adapt to complex, non-linear patterns in the data, and providing more flexible estimates compared to global polynomial or linear methods. However, because LOESS is sensitive to outliers and distributional assumptions within each neighbourhood, validation remain crucial to minimise potential biases or overfitting in the imputation process. Moreover, LOESS might suffer from boundary effects, where the estimation near the edges of the data range is less reliable due to the reduced number of neighbouring data points.

The *k*NN imputation method in the *peatfr* package estimates missing data by identifying the *k* observations that most closely resemble the incomplete record across the observed attributes using a Gower distance. This data imputation method accommodates both continuous and categorical variables without assuming a particular parametric form for the data distribution. However, *k*NN imputation can be computationally intensive for large datasets, because pairwise distances must be recalculated for each imputation. Furthermore, the optimal selection of *k* may exert a substantial impact on imputation accuracy. The mathematical formulation and algorithm for data imputation methods are summarised in Table 2 and Algorithm 1, respectively.

Algorithm 1. The algorithm of data imputation method employed by *peatfr*, including, linear interpolation, cubic spline, LOESS, and *k*NN.

Input.

Step 1. The time series data with missing values.

Step 2. Selected imputation method, including, linear, spline, loess, and knn.

Step 3. Optional method specific parameters.

Process.

Step 4. Check if the method is valid and if the dataset contains

Table 2

Mathematical formulations of the imputation methods implemented in the *peatfr* package, which enable reconstruction of incomplete hydrological and meteorological time series data.

Imputation method	Mathematical equation
Linear interpolation	$y(x) = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0)$ with two known points (x_0, y_0) and (x_1, y_1).
Cubic spline	$y_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$ with continuity in $y, y',$ and y'' at data points and boundary conditions.
LOESS	$y_i(x) = \beta_0 + \beta_1(x - x_i) + \dots + \beta_p(x - x_i)^p$ with $\min_{\beta_0, \beta_1, \dots, \beta_p} \sum_{i=1}^n w_i(x) [y_i - (\beta_0 + \beta_1(x - x_i) + \dots + \beta_p(x - x_i)^p)]^2$ $w_i(x) = \begin{cases} \left[1 - \left(\frac{ x_i - x }{D}\right)^3\right]^3, & x_i - x < D \\ 0, & \text{otherwise} \end{cases}$ D is the distance cutoff derived from the chosen bandwidth.
kNN	$y_i(x) = \text{median} \{y_i i \in N_k(x)\}$ with $N_k(x) = \text{argsmallest}_k \{d_G(x, x_1), \dots, d_G(x, x_n)\}$ $d_G(x, x_i) = \frac{\sum_{m=1}^d w_{i,m} \delta_m(x_m, x_{i,m})}{\sum_{m=1}^d w_{i,m}}$ d_G is the Gower distance.

missing values.

Step 5. Based on the chosen method, apply the appropriate imputation procedure:

- 5.1. Linear: Fill missing values through linear interpolation between known points
- 5.2. Spline: Use cubic spline interpolation to estimate missing values with smooth curvature
- 5.3. LOESS: Apply LOESS smoothing with local regression around each missing point
- 5.4. kNN: Identify k nearest neighbours using Gower distance and impute using their values

Step 6. Return the imputed dataset.

Output.

Step 7. The dataset with missing values filled using the selected method.

2.2. Time series forecasting in *peatfr*

The *peatfr* package employs both stochastic and machine learning approaches to forecast time series data associated with peatland fire risk. The stochastic component uses Autoregressive Integrated Moving Average (ARIMA) model defined as

$$\phi_p(B)(1 - B)^d y_t = c + \theta_q(B)\epsilon_t \quad (1)$$

with $\phi_p(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ and $\theta_q(B) = 1 + \theta_1 B + \dots + \theta_q B^q$. In this formulation B is the backshift operator $B(y_t) = y_{t-1}$, p is the order of AR part, q is the order of MA part, d is the order of differencing, ϕ_i is the AR coefficient, θ_i is the MA coefficient, c is the constant, and ϵ_t is the white noise. ARIMA modelling is a prominent method for time series analysis that captures linear dependence structures and trends within sequential data. ARIMA is widely applied in forecasting due to its robustness, interpretability, ability to handle non stationarity through differencing (Agaj et al., 2024; Kaur et al., 2023; Ridenhour et al., 2017).

In order to stabilise the variance of the data and approximate normality, Box-Cox transformation (Atkinson et al., 2021; Box and Cox, 1964; Proietti and Riani, 2009) is used in the *peatfr* package as follows

$$y_t^{(\lambda)} = \begin{cases} \frac{y_t^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln y_t, & \lambda = 0 \end{cases} \quad (2)$$

where λ is the Box-Cox transformation parameter, y_t is the initial time series data, and $y_t^{(\lambda)}$ is the transformed time series data. We shift the variables in the presence of negative values before applying the Box-Cox transformation because this transformation requires non-negative input data. This transformation improves the effectiveness of model estimation and enhances diagnostic procedures by mitigating issues related to heteroscedasticity and nonlinearity. The Box-Cox transformation and ARIMA modelling form a flexible approach for addressing the complexities encountered in time series data that influences peatland fire. Algorithm 2 summarises the workflow for time series forecasting in the *peatfr* using ARIMA model combined with Box-Cox transformation.

Algorithm 2. The algorithm of time series forecasting through ARIMA model and Box-Cox transformation employed by *peatfr*.

Input.

Step 1. Time series data without missing values y_t .

Step 2. The number of forecasting time h .

Process.

Step 3. Check for negative values in the variable y_t and apply a shifting procedure if negative values are detected.

Step 4. Estimate Box-Cox transformation parameter λ using maximum likelihood.

Step 5. Transform y_t into $y_t^{(\lambda)}$ through the Box-Cox transformation.

Step 6. Stationarity check on $y_t^{(\lambda)}$ with Augmented Dickey-Fuller (ADF) test.

Step 7. Determine ARIMA order (p, d, q) and fit ARIMA model to $y_t^{(\lambda)}$.

Step 8. Conduct residual diagnostics using the Ljung-Box test.

Step 9. Forecast h steps ahead on the transformed scale.

Step 10. Back transform forecasts to original scale.

Output.

Step 11. Final forecast of time series data using ARIMA model.

In addition to the stochastic ARIMA model, the *peatfr* package integrates LSTM and GRU networks to capture nonlinear and long-term dependencies that conventional linear models may overlook. Both LSTM and GRU are advanced forms of Recurrent Neural Network capable of retaining temporal information through their gating mechanisms (Abbasi-mehr and Paki, 2022; Gharehbaghi et al., 2022; Le et al.,

2019). The LSTM architecture includes input, forget, and output gates (Fig. 2a) that regulate the flow of information, allowing the model to learn long-range dependencies and prevent vanishing gradients through the following equations

$$f_t = \sigma(W_{yf}y_t + W_{hf}h_{t-1} + b_f) \quad (3)$$

$$i_t = \sigma(W_{yi}y_t + W_{hi}h_{t-1} + b_i) \quad (4)$$

$$\tilde{c}_t = \tanh(W_{yc}y_t + W_{hc}h_{t-1} + b_c) \quad (5)$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \tilde{c}_t \quad (6)$$

$$o_t = \sigma(W_{yo}y_t + W_{ho}h_{t-1} + b_o) \quad (7)$$

$$h_t = o_t \otimes \tanh(c_t) \quad (8)$$

where σ is the logistic sigmoidal function, $\otimes$ is the element wise multiplication, W_{yf} , W_{hf} , W_{yi} , W_{hi} , W_{yo} , W_{ho} , W_{yc} , W_{hc} are the network weights matrices, b_f , b_i , b_o , b_c are the bias vectors, f_t is the forget gate, i_t is the input gate and o_t is the output gate.

The GRU, a simplified variant of LSTM, merges the input and forget gates into an update gate (Fig. 2b), offering comparable predictive accuracy with reduced computational complexity (Gao et al., 2020; Yao et al., 2023). The mathematical formulation of GRU can be written as

$$r_t = \sigma(W_{yr}y_t + W_{hr}h_{t-1} + b_r) \quad (9)$$

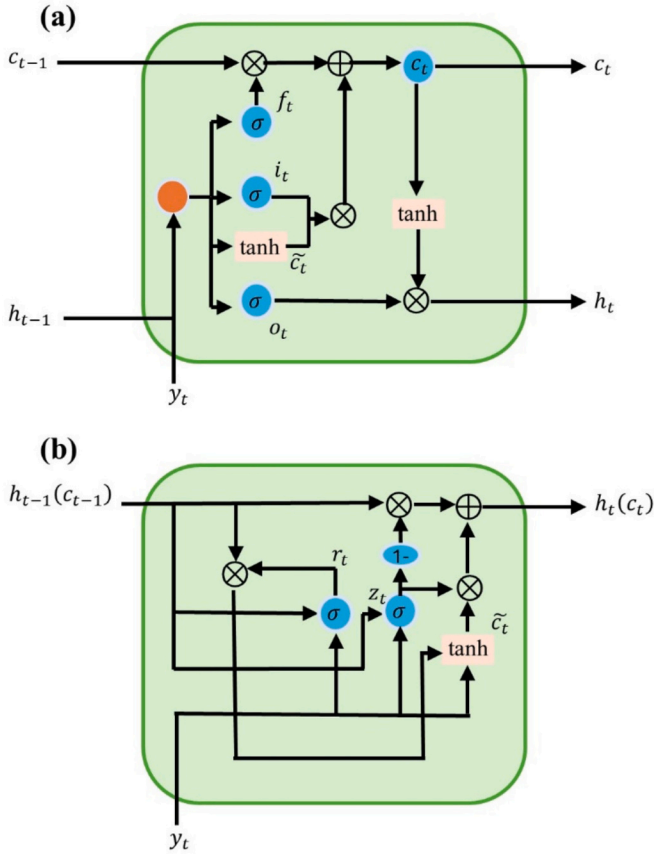


Fig. 2. (a) The architecture of the Long Short-Term Memory (LSTM) network, which includes the input, forget, and output gates. (b) The architecture of the Gated Recurrent Unit (GRU) network, consisting of update and reset gates that simplify the gating mechanism while maintaining the ability to capture sequential relationships and reduce computational complexity compared to LSTM.

$$z_t = \sigma(W_{yz}y_t + W_{hz}h_{t-1} + b_z) \quad (10)$$

$$\tilde{c}_t = \tanh(W_{yc}y_t + W_{hc}(r_t \otimes h_{t-1}) + b_c) \quad (11)$$

$$c_t = (1 - z_t) \otimes c_{t-1} + z_t \otimes \tilde{c}_t \quad (12)$$

$$h_t = c_t \quad (13)$$

where W_{yr} , W_{hr} , W_{yz} , W_{hz} are the network weights matrices, b_r and b_z are the bias vectors, r_t is the update gate, and z_t is the reset gate.

The LSTM and GRU models are implemented using sequential layers in which each neuron processes temporal dependencies through the propagation of hidden states. The input time series are normalised to ensure stable gradient updates and numerical consistency during optimisation. The models are trained using the Adam (Adaptive Moment Estimation) optimisation algorithm, which adaptively adjusts the learning rate for each parameter by computing the first and second moments of the gradients. By adjusting learning rates dynamically, Adam accelerates convergence and provides stable parameter updates even in the presence of noisy or non-stationary gradients. The mean squared error (MSE) is employed as the loss function to quantify prediction accuracy. Algorithm 3 outlines the procedure for time series forecasting in *peatfr* based on LSTM and GRU architectures.

Algorithm 3. The algorithm for time series forecasting in *peatfr* based on LSTM and GRU machine learning methods.

Input.

- Step 1. Time series data without missing values y_t .
- Step 2. The number of forecasting time h .
- Step 3. Define model type: LSTM or GRU.

Process.

- Step 4. Normalise y_t to the range $[0, 1]$ to improve training stability.
 - Step 5. Convert the univariate time series into supervised learning format by creating input-output pairs with a specified look-back window.
 - Step 6. Reshape input data into a 3D tensor to meet the input requirements of recurrent neural networks.
 - Step 7. Define the network architecture.
 - 7.1 For LSTM, include input, hidden, and output layers with memory cells and three gating mechanisms (input, forget, and output gates).
 - 7.2 For GRU, include input, hidden, and output layers with two gating mechanisms (update and reset gates).
 - Step 8. Initialise network weights and compile the model using the Adam optimizer and mean squared error (MSE) as the loss function.
 - Step 9. Train the model using the training dataset, applying early stopping based on validation loss to prevent overfitting.
 - Step 10. Forecast h steps ahead using the trained LSTM or GRU model.
 - Step 11. Apply inverse normalisation to obtain forecasts on the original scale.
- Output.**
- Step 12. Final forecast of time series data using LSTM or GRU model.

2.3. Peatland fire risk index in *peatfr*

The Peat Fire Vulnerability Index (PFVI) (Taufik et al., 2022) is employed by *peatfr* package to analyse fire regimes in tropical peatlands, which can be written as

$$PFVI_t = PFVI_{t-1} + DF_t - RF_t - WTF_t \quad (14)$$

with

$$DF_t = \frac{(300 - PFVI_{t-1})(0.4982e^{(0.0905 \times T_m + 1.6096)} - 4.268 \times 10^{-3})}{1 + 10.88e^{(-0.001736 \times R_0)}} \quad (15)$$

$$RF_t = \begin{cases} (R_t - 5, 1), R_t \geq 5, 1 \text{ mm day}^{-1}, \text{first rainy day} \\ R_t, R_t \geq 5, 1 \text{ mm day}^{-1}, \text{subsequent rainy days} \\ 0, R_t < 5, 1 \text{ mm day}^{-1} \end{cases} \quad (16)$$

$$WTF_t = a_H - b_H \times [(1 - \theta(v))^t \times 300] \quad (17)$$

$$\theta(v) = \left(1 + \left[\frac{v}{\alpha}\right]^n\right)^{-\left(1 - \frac{1}{n}\right)} \quad (18)$$

where DF is the water loss due to evapotranspiration, RF is the rainfall factor, WTF is the water table factor, T_m is the daily maximum air temperature, R_0 is the representative annual rainfall, a_H and b_H are calibration constants, $\theta(v)$ is the volumetric moisture content in the top peat layer at distance v from the surface to water table, α and n are the peat hydraulic properties. The PFVI is initialised at zero, which represents saturated or nearly saturated peat, and increases during dry periods with the maximum value of 300, which suggests severe dryness.

The PFVI is developed based on a dynamic water balance that integrates atmospheric influence with groundwater and soil water retention information. The drought factor DF represents the climatic component by combining maximum temperature and annual rainfall to estimate evaporative water loss, while the rainfall factor RF accounts for effective precipitation. Together, DF and RF quantify the climatic control on peat surface wetness. The water table factor WTF represents the hydrological component by quantifying capillary water contributions from the groundwater table to the unsaturated peat layer. This contribution declines with increasing water table depth, indicating the critical threshold for peat surface drying. The moisture content $\theta(v)$, defined by the van Genuchten parameters α and n , characterises the relationship between groundwater depth and surface moisture. These parameters describe peat water holding capacity and degree of humification, where higher n values indicate degraded peat with limited capillary rise. Through this formulation, the PFVI integrates climatic, hydrological, and soil hydraulic factors into a comprehensive indicator of peat fire vulnerability.

Unlike traditional indices, for example, the Keetch-Byram Drought Index (KBDI), which primarily depend on climatic variables (Ganatsas et al., 2011; Taufik et al., 2015), the PFVI integrates the influence of groundwater dynamics and capillary water movement that affect peat moisture content. Through this approach the PFVI provides a more precise and realistic assessment of peatland fire vulnerability, particularly during dry periods. The index dynamically categorises fire risk into four levels, including low [0 – 75], moderate [75 – 150], high [150 – 225], and extreme [225 – 300], allowing for targeted and timely interventions.

The PFVI parameters a_H , b_H , α and n are calibrated to minimise deviations from the observed drought index that is derived directly from measured peat moisture levels at specific depths. In the *peatfr* package, minimisation process is conducted through Nelder-Mead optimisation method (Takenaga et al., 2023; Wessing, 2019) with the objective function as follows

$$\min_{a_H, b_H, \alpha, n} \sum_t (PFVI_t - DI_{obs})^2, 0 \leq PFVI_t \leq 300 \quad (19)$$

where a_H and b_H are calibration constants, α and n are the peat hydraulic properties, PFVI is the Peat Fire Vulnerability Index, and DI_{obs} is the observed drought index. The Nelder-Mead optimisation method minimises a function with j parameters by comparing the function values at the $j + 1$ vertices of a simplex and replacing the vertex with the highest function value with another point using reflection, expansion, and contraction steps. If neither reflection nor contraction provides improvement, the algorithm shrinks the simplex toward the current best point. An important advantage of this method is that it does not require derivative calculations, which suitable for functions that are non-differentiable or exhibit discontinuities. The algorithm to determine

peatland fire index in *peatfr* is summarised in Algorithm 4.

Algorithm 4. The algorithm to calculate peatland fire index in *peatfr* through Peat Fire Vulnerability Index (PFVI) and Nelder-Mead optimisation method.

Input.

- Step 1. Hydrological data, including, water table and soil moisture.
- Step 2. Meteorological data, including, rainfall and air temperature.

Process

- Step 3. Calculate volumetric moisture content $\theta(v)$.
- Step 4. Calculate water table factor WTF_t .
- Step 5. Calculate rainfall factor RF_t .
- Step 6. Calculate the drought factor DF_t .
- Step 7. Calculate PFVI and apply the constraint between 0 until 300.
- Step 8. Calibration through Nelder-Mead optimisation method with the objective function is designed to minimise the deviations between the calculated PFVI with the observed drought index.

- 8.1. Initialise simplex of parameters combinations a_H , b_H , α and n .
- 8.2. Iteratively apply reflection, expansion, contraction, and shrink.
- 8.3. Stop when change in objective function is below threshold.

Output.

- Step 9. Optimised parameters a_H , b_H , α and n .
- Step 10. Time series of PFVI.

2.4. Outputs visualisation in *peatfr*

The *peatfr* offers an automated visualisation feature that enables users to interpret the outputs generated from each process within the package. Visualisation plays an essential role in helping to present complex data analysis or modelling processes in a more accessible and understandable form. The visualisation of data imputation process produces figures of the completed time series dataset with different colour between the original and imputed data. This distinction allows users to identify which portions of the dataset have been estimated due to missing values. By providing this visualisation, the *peatfr* not only ensures traceability in the imputation process but also increases user confidence in the reliability of the dataset. The ability to directly observe the location and proportion of imputed values helps users assess the quality of their data before proceeding to the forecasting analysis.

The visualisations of forecasting outputs are also presented with distinct colour schemes to differentiate between historical data and future predictions. This visual distinction helps users to track the transition from observed values to forecasted values within the same plot. This feature is particularly valuable for supporting decision making processes in tropical peatland fire risk management because it provides a clear visual representation of water table, soil moisture, rainfall, temperature, and PFVI together with the anticipated future trends. Based on these visualisation outputs, users can identify periods of increased fire risk and prepare appropriate mitigation strategies accordingly.

3. Example application of *peatfr*

This section demonstrates the practical usage of the *peatfr* package to analyse the fire risk of tropical peatland based on the conceptual framework shown in Fig. 1. The data employed in this study are the average daily of water table, soil moisture, rainfall, and daily maximum of air temperature from tropical peatland in Sabangau, Central Kalimantan, Indonesia between 1 April 2023 and 9 October 2023 (Fig. 3). These data provide a comprehensive information of hydrological and meteorological conditions, which are crucial in understanding and predicting peatland fire dynamics (S. Mezbahuddin et al., 2023; Taufik et al., 2022; Widyastuti et al., 2021). The data were obtained from the Peatland and Mangrove Restoration Agency (BRGM), and Meteorology, Climatology, and Geophysical Agency (BMKG) of the Republic of Indonesia, both of which operate remote sensing and ground based observation platforms. The BRGM satellite monitoring system provides

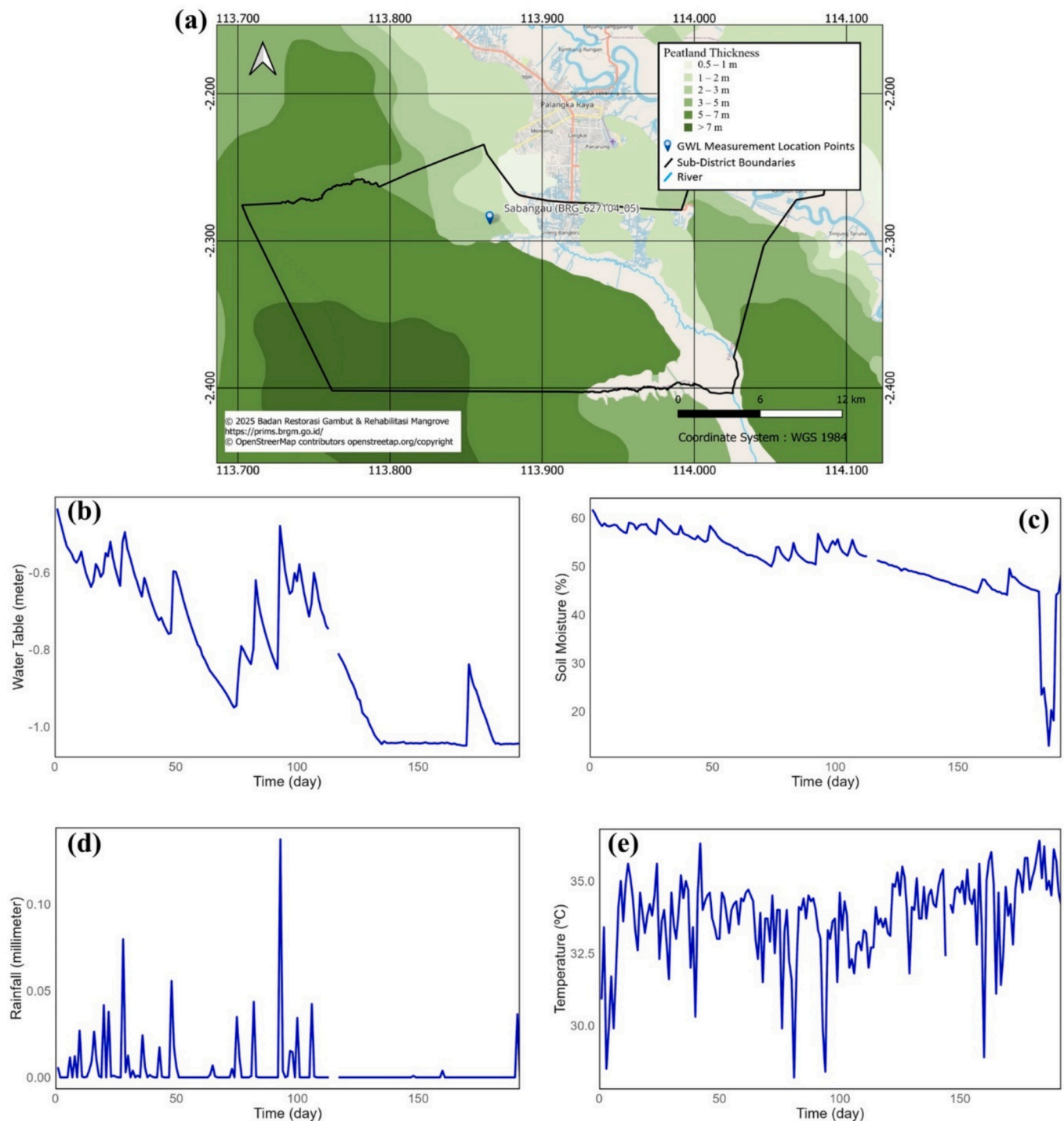


Fig. 3. (a) The peatland map of Sabangau in Central Kalimantan, Indonesia serves as the demonstration site for the application of the *peatfr* package. The initial time series data of (b) water table, (c) soil moisture, (d) rainfall, and (e) temperature obtained from satellite observation of Peatland and Mangrove Restoration Agency (BRGM) and Meteorology, Climatology, and Geophysical Agency (BMKG) in Sabangau. Missing values occurs on days 114 to 116 for water table, soil moisture, and rainfall, and on day 145 for temperature.

near real time water table, soil moisture, and rainfall data across peatland regions, while BMKG provides continuously updated temperature data derived from weather stations and satellite-based climate observation networks.

The *peatfr* introduces a function designed to simulate the entire peatland fire prediction process comprehensively through *autopeatfr* () with the code as follows.

```
instal.packages("devtools")

devtools::install_github("mellygsln/peatfr")

library(peatfr)
```

When the ARIMA model is employed for time series forecasting, the

corresponding function call in *peatfr* is.

```
firerisk <- autopeatfr(WT, SM, Rf, Temp, imputation = "knn",
model = "ARIMA", h = 4, R0 = 2700, dt = 1)
```

Alternatively, when the LSTM or GRU model is used for time series forecasting, the function call is modified as follows

```
firerisk <- autopeatfr(WT, SM, Rf, Temp, imputation = "knn",
model = c("LSTM", "GRU"), look_back = 12, layer_units = 32, epochs
= 100, h = 4, R0 = 2700, dt = 1)
```

In these codes, *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, *imputation* is the imputation

```
library(peatfr)
```

```
imputation <- knn_imputation(WT, SM, Rf, Temp, k = 5)
```

tation method, *model* is the time series forecasting method, *look_back* is the time window, *layer_units* is the number of memory cells, *epochs* is the complete training of the training data set through the neural network, *h* is the number of forecasting time, *R₀* is the representative annual rainfall in the location, and *dt* is the time interval for each observation.

The *peatfr* is also structured to be modular, resulting in the flexibility of its application. Each function within the framework corresponds to a specific stage of the prediction process and can operate independently, enabling integration with other models. This modularity enhances adaptability, allowing for refinement or expansion of individual components without disrupting the overall predictive framework.

3.1. Imputation for missing data

We found that the water table, soil moisture, rainfall, and air temperature in our dataset contain missing observation (Fig. 3). The missing data could lead to the bias results and compromise the performance of

time series analysis. To overcome this issue, we employ *kNN* imputation method with Gower distance to quantify similarity among data points (Tuerhong and Kim, 2014) through the function *knn_imputation()*.

where *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, and *k* is the number of neighbours. By filling in the missing values with the smallest dissimilarities data, we could minimise distortions in the data and enhance the overall performance of the time series predictions.

Fig. 4 shows the imputed data for the water table, soil moisture, rainfall, and air temperature. A comparison of the mean, median, and standard deviation between imputed and non-imputed data sets (Table 3), shows minimal discrepancies. Furthermore, examining box-plots of these variables before and after imputation indicates negligible shifts in the quartiles and outlier positions (Fig. 5). The consistency in

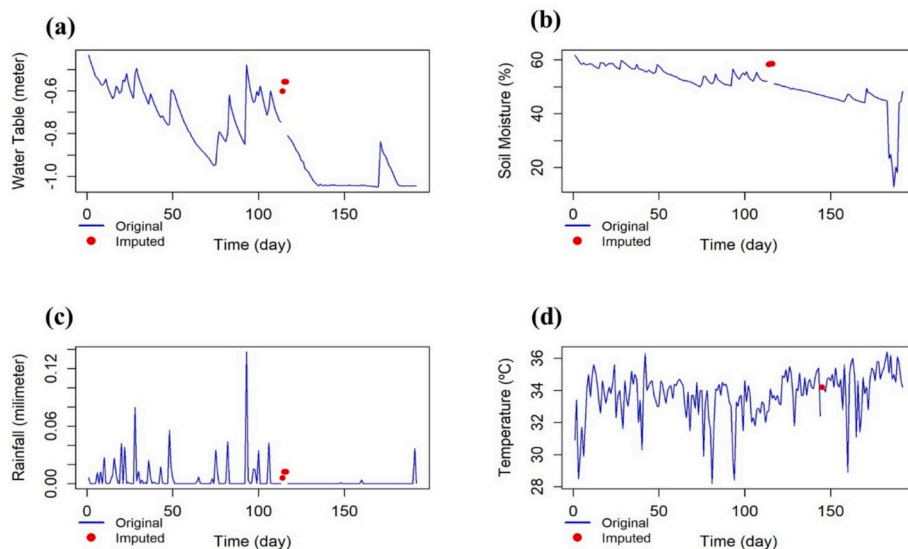


Fig. 4. The results of *kNN* imputation for (a) water table, (b) soil moisture, (c) rainfall, and (d) air temperature.

Table 3

The mean, median, and standard deviation of water table, soil moisture, rainfall, and air temperature before and after imputation process.

	Water table		Soil moisture		Rainfall		Air temperature	
	Before	After	Before	After	Before	After	Before	After
Mean	−0.812	−0.808	51.039	51.156	0.004	0.004	33.743	33.746
Median	−0.817	−0.815	52.037	52.086	0.000	0.000	34.000	34.000
Standard deviation	0.187	0.188	7.304	7.306	0.014	0.014	1.484	1.480

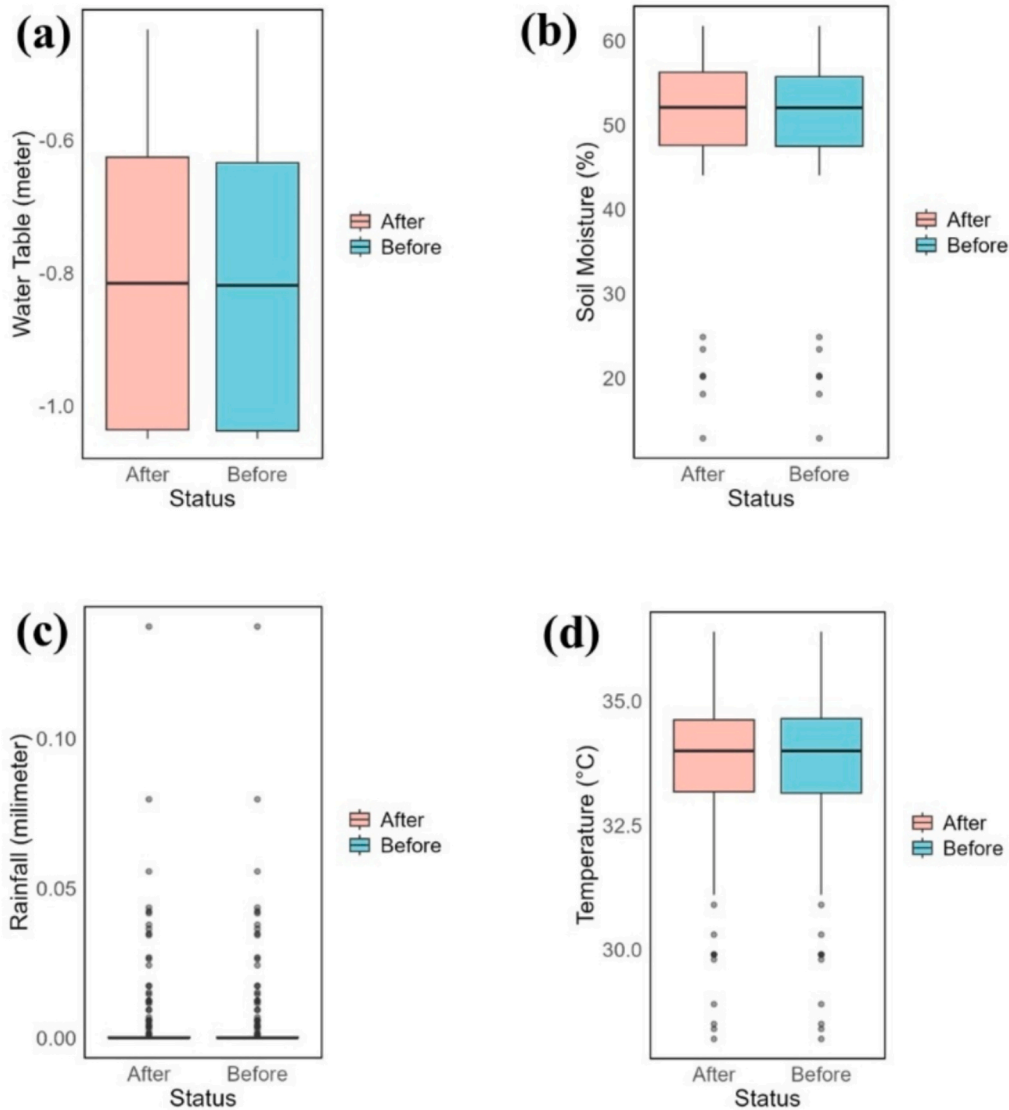


Fig. 5. The boxplot comparison between original and imputed data for (a) water table, (b) soil moisture, (c) rainfall, and (d) air temperature to analyse the changes in quartiles and outlier positions.

both descriptive statistics and distributional characteristics suggests that the imputation process has effectively preserved the original structure of the data without introducing bias.

3.2. Forecasting tropical peatland fire risk

Time series forecasting in *peatfr* can be performed using two complementary categories of models, comprising a stochastic method

represented by the ARIMA model and a more advanced machine learning approaches that include the LSTM and GRU networks. These methods enable the modelling of both linear and nonlinear temporal dependencies in hydrological and meteorological variables that influence peatland fire risk, including, water table, soil moisture, rainfall, and air temperature.

The forecasting of these variables using ARIMA model is implemented through the function `autopredictarima()`, which automates

the process of identifying and fitting the optimal ARIMA model by minimising prediction errors.

```
library(peatfr)

timeseriespredict <- autopredictarima(WT, SM, Rf, Temp, h = 4)
```

where *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, and *h* is the number of forecasting time. Rather than manually specifying model parameters through autocorrelation function (ACF) and partial autocorrelation function (PACF), `autopredictarima()` tests various combinations, ultimately identifying the configuration that achieves the lowest error metrics. Moreover, it also addresses problems related to the non-constant variance and heteroscedasticity in the data using Box-Cox transformation.

We forecasted values of water table, soil moisture, rainfall, and air temperature for the subsequent four-day period with ARIMA, as shown in Fig. 6. The short-term projections offer immediate insights into the

```
library(peatfr)

timeseriespredict <- autopredictgru(WT, SM, Rf, Temp, h = 4,
look_back = 12, gru_layer_units = 32, epochs = 100)
```

trends and fluctuations of these data. In the next four days, the water table depth and rainfall are relatively stable with the value around 1.05 m and 0 mm respectively. Contrastingly, soil moisture decreases from about 45 % to 39 % and air temperature increases from around 34.5 °C to 35.5 °C. Furthermore, the evaluation indices for ARIMA model, including mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE), for each variable are summarised in Table 4.

The forecasting process can also be conducted through the function `autopredictlstm()`, which constructs and trains a LSTM network to capture nonlinear dependencies within the hydrological and meteorological time series data.

```
library(peatfr)

timeseriespredict <- autopredictlstm(WT, SM, Rf, Temp, h = 4,
look_back = 12, lstm_layer_units = 32, epochs = 100)
```

where *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, *h* is the number of forecasting time, *look_back* is the time window, *lstm_layer_units* is the number of memory cells in LSTM model, *epochs* is the complete training of the

```
library(peatfr)

pfvi <- firepredict(WT, SM, Rf, Temp, R0 = 2700, dt = 1, h = 4)
```

training data set through the neural network. This function allows users

to specify model parameters, for example, the number of layer units, look-back period, and epochs, producing flexible model configuration

for various data characteristics.

We forecasted the values of water table, soil moisture, rainfall, and air temperature for the subsequent four-day period using the LSTM model, as illustrated in Fig. 7. Over the four-day horizon, the water table depth remains nearly constant at approximately 1.03 m, and rainfall slightly increases from 0 mm to 2.8×10^{-3} mm. In addition, soil moisture gradually raises from approximately 34 % to 35 %, while air temperature shows a slight decline from 34.6 °C to 34.4 °C. The performance of the LSTM model is evaluated by statistical indicators, including MSE, RMSE, and MAE, summarised in Table 5.

Similarly, the function `autopredictgru()` performs time series forecasting using a GRU architecture, a simplified variant of LSTM that maintains comparable predictive performance with reduced computational complexity.

where *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, *h* is the number of forecasting time, *look_back* is the time window, *gru_layer_units* is the number of memory cells in GRU model, *epochs* is the complete training of the training data set through the neural network.

We forecasted the values of water table, soil moisture, rainfall, and air temperature for the next four-day period using the GRU model, as presented in Fig. 8. Throughout the forecasting period, the water table depth remains stable at around 1.04 m, while rainfall exhibits a minor increase from 0 mm to approximately 3.9×10^{-3} mm. Moreover, soil moisture shows a modest rise from 42 % to 45 %, and air temperature

slightly decreases from 34.1 °C to 33.8 °C. The predictive performance of the GRU model, evaluated through MSE, RMSE, and MAE, summarised in Table 6.

The PFVI is calculated using `firepredict()` function through the following code.

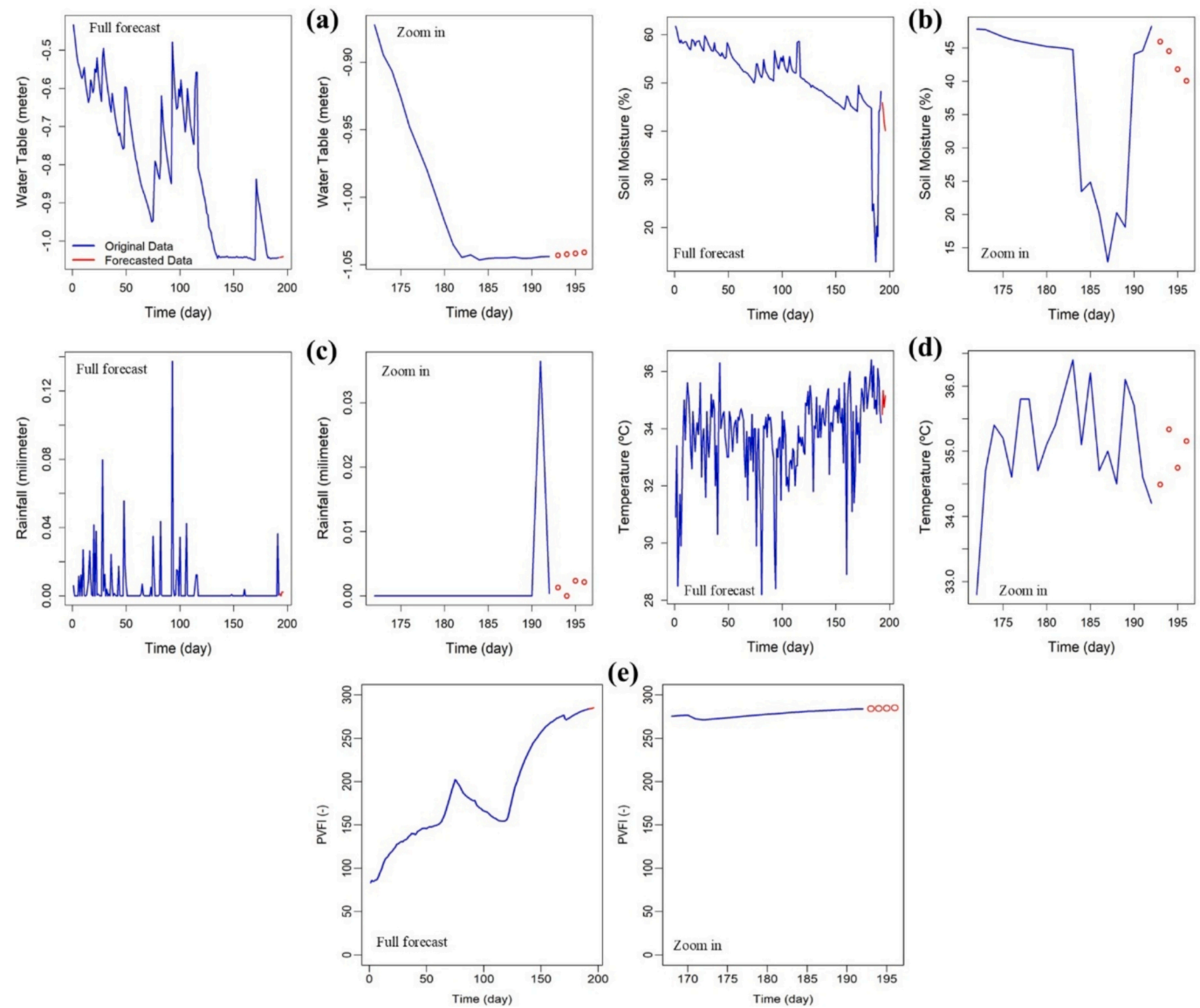


Fig. 6. The forecasting results of (a) water table, (b) soil moisture, (c) rainfall, (d) air temperature, and (e) PFVI for the next four days with ARIMA model.

Table 4
The evaluation indices, including MSE, RMSE, and MAE, for water table, soil moisture, rainfall, and air temperature with ARIMA model.

Evaluation indices	Water table	Soil moisture	Rainfall	Air temperature
MSE	3.329×10^{-3}	3350.624	5.115×10^{-5}	3.254
RMSE	5.769×10^{-2}	57.884	7.152×10^{-3}	1.804
MAE	3.579×10^{-2}	56.521	5.763×10^{-3}	1.569

where *WT* is the time series data of water table position, *SM* is the time series data of soil moisture, *Rf* is the time series data of rainfall, *Temp* is the time series data of air temperature, *R₀* is the representative annual rainfall in the location, *dt* is the time interval for each observation, and *h* is the number of forecasting time. The value of PFVI was distributed across three classes, including moderate, high, and extreme, with the minimum and maximum value of about 90 and 285 with ARIMA model (Fig. 6e), 83 and 293 with LSTM model (Fig. 7e), and 83 and 284 with GRU model (Fig. 8e). This condition shows that hydrological and

meteorological conditions can vary significantly over time and influences the overall fire vulnerability profile. Moreover, the forecasted value of PFVI with ARIMA, LSTM, and GRU models in the next four days indicates that PFVI remains in the extreme class, emphasising a persistent high risk condition that might require immediate and coordinated interventions.

3.3. Sensitivity analysis of PFVI parameters

To examine the stability of the PFVI parameters estimated through the Nelder–Mead optimisation method, all four input variables, including, water table, soil moisture, rainfall, and air temperature were simultaneously perturbed within 10% of their original magnitudes in 2% increments. This simultaneous adjustment allows the collective influence of hydrological and meteorological variability to be tested while maintaining the same optimisation configuration. For each perturbation level, the PFVI parameters were optimised, and the resulting parameter sets were compared with the baseline values to evaluate the consistency of the calibration under gradually changing input conditions.

The results show that all PFVI parameters varied systematically but remained within a narrow and physically consistent range, confirming

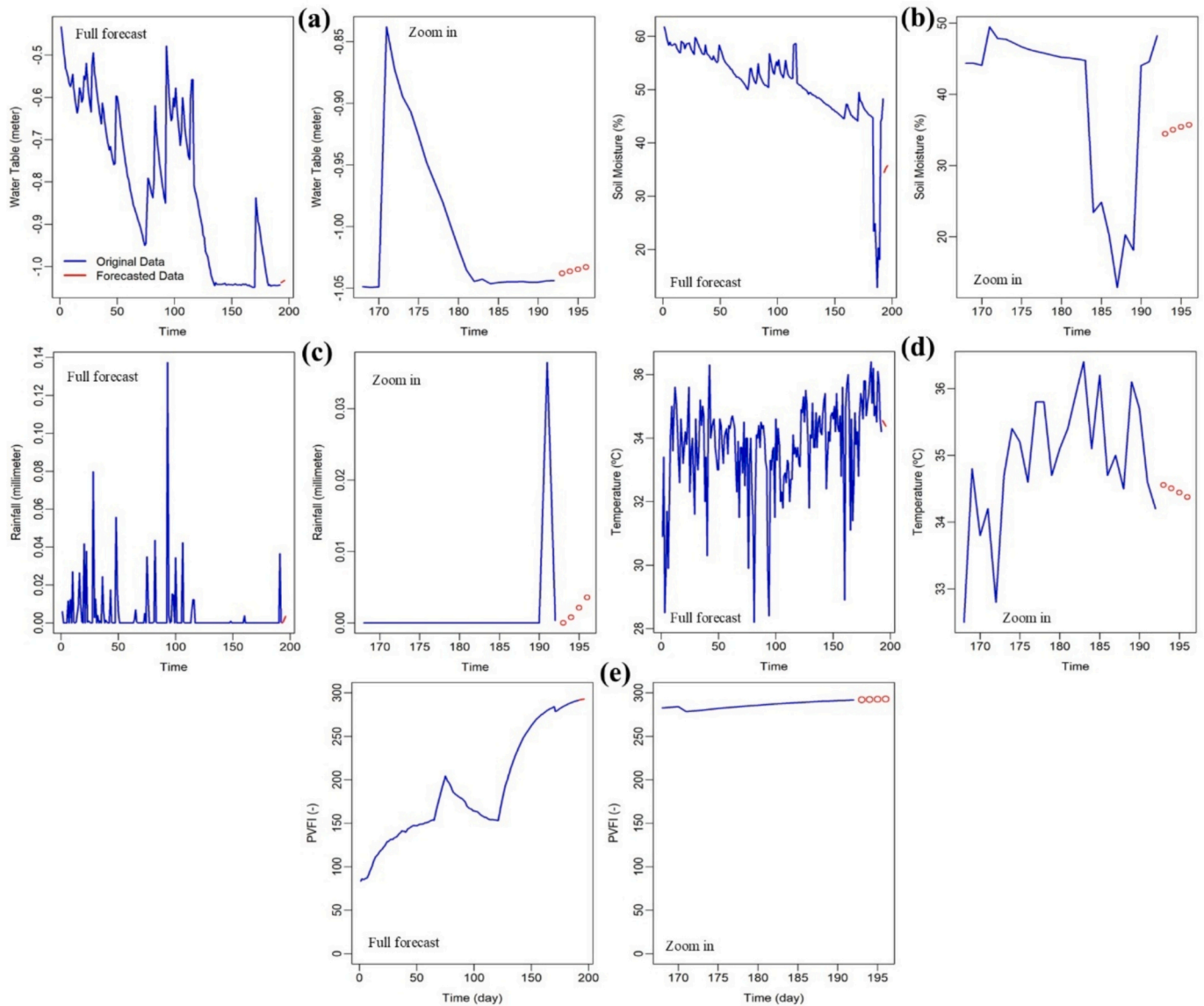


Fig. 7. The forecasting results of (a) water table, (b) soil moisture, (c) rainfall, (d) air temperature, and (e) PFVI for the next four days with LSTM model.

Table 5
The evaluation indices, including MSE, RMSE, and MAE, for water table, soil moisture, rainfall, and air temperature with LSTM model.

Evaluation indices	Water table	Soil moisture	Rainfall	Air temperature
MSE	3.291×10^{-3}	134.211	5.429×10^{-5}	1.381
RMSE	5.736×10^{-2}	11.585	7.368×10^{-3}	1.175
MAE	4.003×10^{-2}	7.463	4.997×10^{-3}	9.972×10^{-1}

the robustness of the optimisation process (Table 7). As the inputs increased by 10%, the a_H rose from 6.501 to 12.874, the b_H increased modestly from 0.020 to 0.030, and the α increased slightly from 0.859 to 0.932. In contrast, the n decreased from 55.342 to 18.264, reflecting its inverse relationship with the hydrological and meteorological inputs. Despite these variations in magnitude, the parameters maintained consistent trends, indicating that the optimisation converged toward a stable parameter region. These findings demonstrate that the PFVI optimisation remains numerically stable and resilient under

simultaneous perturbations of the input variables, supporting the reliability of the model across diverse environmental settings.

4. Discussion

This paper developed an R package *peatfr* to forecast tropical peatland fire phenomenon through the stochastic, machine learning, and optimisation methods. The *peatfr* is straightforward in predicting the tropical peatland fire because it only requires the input time series data of water table, soil moisture, rainfall, and air temperature. In contrast to the software proposed by Mezbahuddin et al. (2023), which depends on the external weather forecasts, the *peatfr* operates as self-contained system. This independence enhances its stability and simplicity, particularly related to the data integration process.

The *peatfr* automatically selects the best ARIMA stochastic model for each time series data and generates forecasts based on these models. Although ARIMA is not a novel method, it remains a highly acclaimed and efficient forecasting model, particularly within the field of hydrology (Birylo et al., 2017; Mirzavand and Ghazavi, 2015; Valipour et al., 2013). For example, Agaj et al. (2024) demonstrated the effectiveness of the ARIMA model in accurately representing and predicting river water

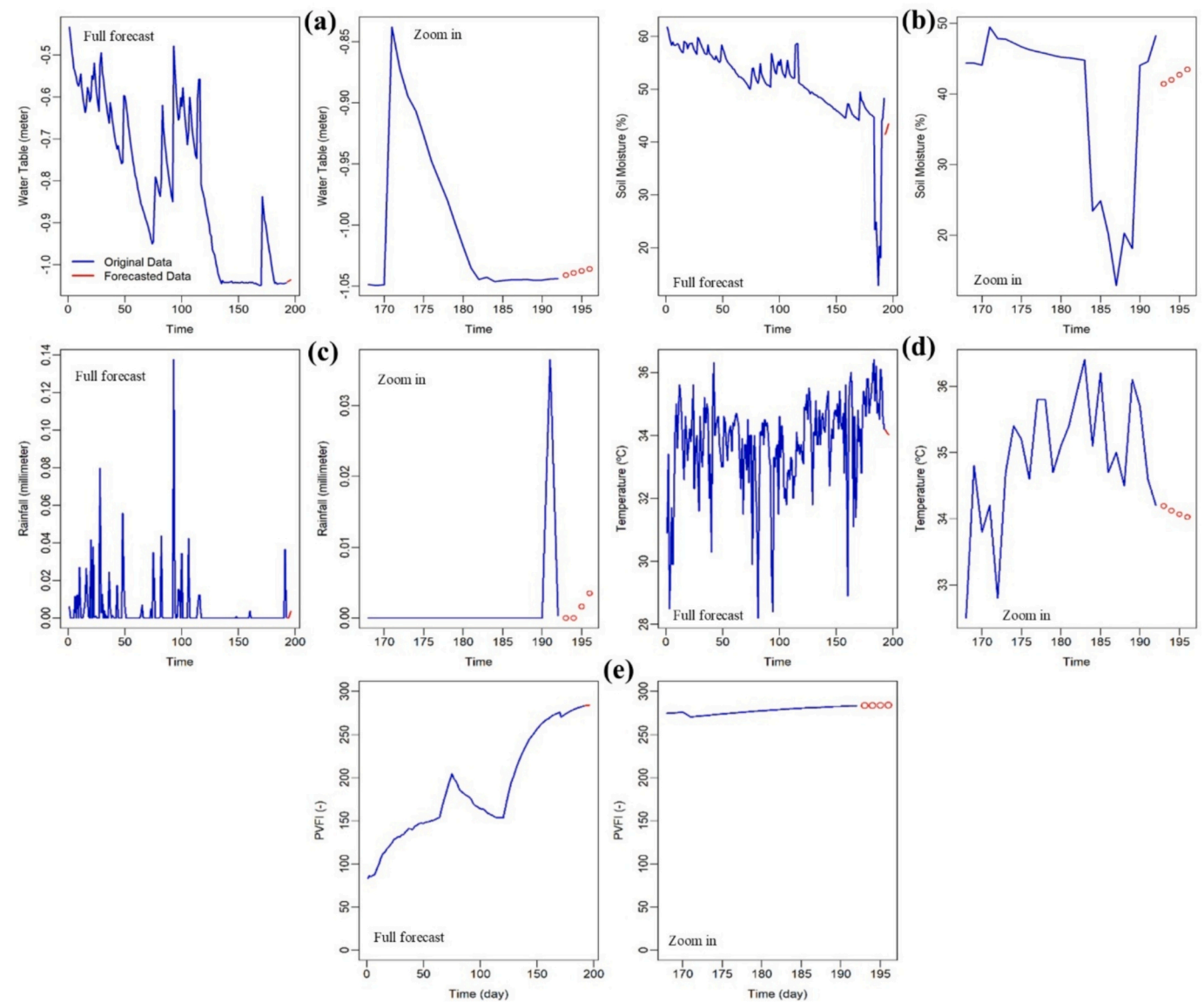


Fig. 8. The forecasting results of (a) water table, (b) soil moisture, (c) rainfall, (d) air temperature, and (e) PFVI for the next four days with GRU model.

Table 6
The evaluation indices, including MSE, RMSE, and MAE, for water table, soil moisture, rainfall, and air temperature with GRU model.

Evaluation indices	Water table	Soil moisture	Rainfall	Air temperature
MSE	2.156×10^{-3}	90.451	5.228×10^{-5}	1.415
RMSE	4.643×10^{-2}	9.510	7.231×10^{-3}	1.189
MAE	2.279×10^{-2}	6.75	4.627×10^{-3}	9.659×10^{-1}

levels. Furthermore, the flexibility and adaptability of ARIMA to be integrated with the other methods also demonstrates its the advantageous and versatility (Phan and Nguyen, 2020). In addition to ARIMA, *peatfr* also incorporates Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models, which represent advanced machine learning architectures capable of capturing nonlinear and long-term temporal dependencies in time series data (Abbasimehr and Paki, 2022; Gao et al., 2020; Le et al., 2019; Yao et al., 2023). These models employ internal gating mechanisms that regulate

the flow of information, allowing them to retain memory of past states and effectively model delayed feedback processes of important hydrological and meteorological variables, including, water table, soil moisture, rainfall, and temperature. By combining stochastic and machine learning approaches, *peatfr* provides a balanced and adaptive framework for improving the accuracy of peatland fire risk prediction. The parameters in *peatfr* are obtained from optimisation process through Nelder-Mead method (Takenaga et al., 2023; Wessing, 2019). By automatically converging on optimal parameter values, this process minimises the risk of bias from user and enhances the overall robustness of the model. This automated optimisation framework from *peatfr* offers a more reliable and objective approach compared to PeatFire software that requires global and local sensitivity analysis with full factorial experiment to obtain the best combinations of parameters (Widyastuti et al., 2021). It not only simplifies the workflow by reducing the need for extensive adjustments but also improves the model predictive accuracy by consistently adapting to the specific characteristics of the dataset. This methodological advantage provides a more effective tool for predicting tropical peatland fire.

Table 7

The sensitivity results of PFVI parameters with different perturbation of input variables.

PFVI parameters	Perturbation of input variables					
	Baseline	2%	4 %	6 %	8 %	10%
a_H	6.501	7.541	8.680	9.934	11.331	12.874
b_H	0.020	0.021	0.023	0.025	0.027	0.030
n	55.342	40.218	32.486	26.672	21.906	18.264
α	0.859	0.874	0.888	0.903	0.917	0.932

4.1. Comparison between stochastic and machine learning methods

The comparison between stochastic and machine learning methods indicate that all models provide satisfactory results, but with notable differences in predictive accuracy across variables. For soil moisture and air temperature, both LSTM and GRU achieved significantly lower error values than ARIMA, indicating their superior capability in representing nonlinear dynamics. The RMSE of soil moisture predictions decreased from 57.884 in ARIMA to 11.585 in LSTM and 9.510 in GRU, while the MAE declined from 56.521 to 7.463 and 6.750, respectively. Similarly, air temperature predictions improved, with RMSE decreasing from 1.804 (ARIMA) to 1.175 (LSTM) and 1.189 (GRU). These results demonstrate that the recurrent neural network architectures, including LSTM and GRU, could better accommodate temporal variability and nonlinear dynamics of time series data.

Conversely, for water table depth and rainfall, the differences among models were less pronounced. The RMSE values for water table prediction were 0.0577 for ARIMA, 0.0574 for LSTM, and 0.0464 for GRU, while rainfall exhibited similar levels of error across all models. This indicates that for time series with relatively stable or recurring seasonal patterns, particularly those found in hydrological variables, the ARIMA model can achieve comparable accuracy while maintaining computational efficiency.

These findings highlight an essential trade-off between interpretability and predictive performance of the proposed methods. The ARIMA stochastic model provides straightforward and interpretable results that facilitate the understanding of the underlying process (Agaj et al., 2024; Yan et al., 2022). Contrastingly, the LSTM and GRU models demonstrate higher predictive capability for nonlinear and time dependent variables (Gao et al., 2020; Gharehbaghi et al., 2022; Yao et al., 2023). The *peatfr* package incorporates both approaches, allowing users to select either the stochastic ARIMA model or the machine learning methods (LSTM and GRU) according to their data characteristics and analytical objectives. By providing this flexibility, *peatfr* combines the interpretability of ARIMA with the adaptability of machine learning, offering a robust and versatile framework for peatland monitoring and fire vulnerability prediction.

4.2. Forecasting capability of *peatfr*

The *peatfr* employs Peat Fire Vulnerability Index (PFVI) to calculate the drought and fire risk in tropical peatland, which incorporates groundwater, soil water retention, and weather information (Taufik et al., 2022). Compared to the other fire risk index that focus on the weather or groundwater factors (Chelli et al., 2015; Ganatsas et al., 2011; Garcia-Prats et al., 2015; Taufik et al., 2015), PFVI involves a more comprehensive variables that significantly influence tropical peatland fire. For example, the peat-dry-index, which is used by PeatFire software (Widyastuti et al., 2021), is affected solely by water table depth to determine the drought condition and fire risk in tropical peatland. However, the water table depth only determines the lower boundary of peatland ecosystem that susceptible to fire. The upper boundary is controlled by the peat capillary pressure and water retention (Rudiyanto et al., 2020; Taufik et al., 2019), which influence the movement of water in unsaturated zone above the water table and the ability to retain the

water within the pore space, respectively. As a consequence, peat-dry-index might omit the critical factor to forecast the occurrence of tropical peatland fire and reduce the predictive capability of the PeatFire software.

The PFVI predictions generated by *peatfr* using ARIMA, LSTM, and GRU models for the subsequent four days, started from 10 October 2023 until 13 October 2023, are in the extreme classification ($PFVI \geq 225$), indicating a high possibility of tropical peatland fire. This classification suggests that combination of hydrological and meteorological factors, including, deep water table position, reduced soil moisture, limited rainfall, and high temperature (Figs. 6–8), created favourable conditions to the occurrence of tropical peatland fires. A lower water table accelerates the drying of peat and increases the risk of fire events (Afira and Wijayanto, 2022; Taufik et al., 2017). Additionally, the decline in water table leads to the soil moisture reduction, which directly affects the availability of fuel for combustion (Hrysiewicz et al., 2023; Nelson et al., 2021). These conditions are further intensified by the combination of low rainfall and high temperature, which support the likelihood of ignition and fire spread (Dastour et al., 2024; Leng et al., 2018; Turetsky et al., 2015). Therefore, *peatfr* anticipates that the fire events will occur around the observation station in Sabangau.

The accuracy of this prediction was validated by the actual occurrence of fire hotspots in the observation station surroundings obtained from NASA Fire Information for Resource Management System (FIRMS) during the period of 10–13 October 2023 (Fig. 9). The dataset was derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) Collection 6 (MCD14ML), which provides daily, near-real-time active fire locations at a 1 km spatial resolution. Each hotspot represents the centroid of a 1 km² grid cell containing at least one active fire, detected using the MODIS MOD14/MYD14 Fire and Thermal Anomalies algorithm (Giglio et al., 2003) based on brightness temperature data from the Terra and Aqua satellites.

We compared the predicted extreme classification generated by *peatfr* with the spatial distribution of actual MODIS fire hotspots. The results demonstrated a strong correspondence between predicted and observed fire occurrences, confirming the model capability to capture the underlying hydrological and meteorological drivers of peat fire events. This agreement indicates a high level of predictive credibility and supports the reliability of *peatfr* as an early warning and risk assessment tool.

4.3. Effective fire mitigation strategies with *peatfr*

The *peatfr* enhances fire mitigation strategies by providing early warnings based on hydrological and meteorological conditions. By predicting the likelihood of tropical peatland fires, stakeholders can take proactive measures to prevent and control fire outbreaks (Horton et al., 2022; Murdiyarso et al., 2019). This includes mobilising firefighting teams, deploying water resources, and reinforcing firebreaks in high-risk areas before fires escalate. Additionally, the predictions from *peatfr* might be employed by local authorities or land managers to impose temporary land-use restrictions, for example, controlled burning bans, during periods of extreme fire risk.

Beyond emergency response, *peatfr* supports long-term fire prevention strategies. By identifying patterns in fire-prone conditions, policy-makers can develop sustainable water management plans to maintain peatland moisture levels, which reduces susceptibility to ignition (Fawzi et al., 2024). This tool also aids in community preparedness by raising awareness and guiding local fire prevention initiatives (Purnomo et al., 2024). Integrating *peatfr* into peatland management frameworks (Monteverde et al., 2022) ensures a data driven approach to fire risk reduction, resulting in the protection of both ecosystems and infrastructures from the devastating impacts of uncontrolled peatland fires.

4.4. Limitation and future development

The *peatfr* employs Peat Fire Vulnerability Index (PFVI), which is specifically developed and tested for fire occurrence in tropical peatland as a model for fire danger index (Taufik et al., 2022). PFVI emphasises the hydrological and meteorological variables that exert the most direct control over tropical peat surface wetness, including water table, soil moisture, rainfall, and temperature. These variables represent the essential physical drivers of peat combustion because the ignition and spread of peat fires are primarily governed by subsurface dryness and the depletion of near surface moisture (Harrison et al., 2024; Koupaei-Abyazani et al., 2024; S. Mezbahuddin et al., 2023; Taufik et al., 2017). The importance of water table position for fire occurrence in tropical peatland has been shown through the simulations and field observations. For example, Wösten et al. (2008) employed the groundwater model and satellite image to analyse the susceptibility of peat fire in Central Kalimantan, Indonesia. Furthermore, the field observation from Konecny et al. (2016) showed that extensive drainage of peatlands leads to persistently deep water table position, resulting in frequent fire occurrences on degraded peat areas. While hydrological factors play a dominant role in controlling peat fire vulnerability, meteorological conditions also contribute significantly by regulating peat moisture and ignition thresholds. Leng et al. (2018) indicated that changes in temperature and rainfall are the main variables influencing the frequency and intensity of fires in tropical peatlands. However, it is recognised that broader climate variables, fuel conditions, and vegetation composition might provide crucial roles in fire initiation and propagation. Widely recognised indices, for example, the Fire Weather Index (FWI) (Chelli et al., 2015; Groot et al., 2007) and the Forest Fire Danger Index (FFDI) (Sanabria et al., 2013) encapsulate these additional factors by integrating wind, humidity, and vegetation characteristics. These fire indices could be integrated into the *peatfr* package through the coupling of PFVI outputs with FWI and FFDI using both statistical and machine learning approaches to create a dynamic and data driven framework.

The *peatfr* employs ARIMA model to predict the future value of water table position through the function `autopredictarima()`. As a result, it ignores the influence of spatial variability, for example, physical properties and topography, on water table position. Mahdiyasa et al. (2024) showed that the water table exhibits not only temporal but also spatial variability. The water table is located in the deeper position at the margin compared to the centre because of the changes in hydraulic conductivity, bulk density, and hydraulic gradient (Baird et al., 2008; Lapen et al., 2005; Lewis et al., 2012). To overcome the limitation of

neglecting spatial effects, the Generalized Space Time Autoregressive (GSTAR) model could be integrated into the package, because it allows the autoregressive parameters to vary across locations (Mukhaiyar et al., 2024a, 2024b). This flexibility enables the model to account for local differences in hydrological behaviour, capturing the specific characteristics of different regions within the peatland. Consequently, the enhanced model could provide more accurate and spatially resolved water table predictions.

The *peatfr* has the potential to be expanded into an interactive dashboard for real time tropical peatland fire monitoring and decision making. The dashboard could provide stakeholders with comprehensive insights into fire risk levels, hydrological and meteorological conditions, and necessary mitigation measures by integrating *peatfr* predictive capabilities with a user-friendly visual interface. Furthermore, it could display dynamic maps of predicted fire prone areas, real-time updates on PFVI, and trends in important environmental variables, including, water table depth, soil moisture, rainfall, and temperature. By transforming *peatfr* into a dashboard, the system might serve as an essential tool for proactive fire management (Purnomo et al., 2024), improving both short-term emergency response and long-term tropical peatland conservation efforts (Horton et al., 2022).

5. Conclusion

The R package *peatfr* offer comprehensive framework for forecasting tropical peatland fire with three main functions, including, data imputation process, time series forecasting, and fire risk calculation. The data imputation model in *peatfr* consists of linear interpolation, spline, locally estimated scatterplot smoothing (LOESS), and *k*-nearest neighbour (*k*NN), which can be employed to estimate the missing values of time series data of water table, soil moisture, rainfall, and air temperature. The forecasting process integrates both stochastic and machine learning approaches through the Autoregressive Integrated Moving Average (ARIMA), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) models to capture time dependent dynamics in hydrological and meteorological variables. To classify the drought and fire risk of tropical peatland, *peatfr* uses Peat Fire Vulnerability Index (PFVI) that incorporating groundwater, soil water retention, and weather information. The PFVI parameters are calculated automatically based on the input data through Nelder-Mead optimisation, which ensures the reliability of this package. The capability of *peatfr* to predict the tropical peatland fire events plays a crucial role in guiding and implementing effective fire mitigation strategies, which could reduce the risk of extensive fire damage. Finally, this package may be of assistance to develop an efficient early warning system of tropical peatland fire.

CRedit authorship contribution statement

Adilan W. Mahdiyasa: Writing – review & editing, Writing – original draft, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Melly Melly:** Writing – review & editing, Software, Methodology. **Udjianna S. Pasaribu:** Writing – review & editing, Methodology. **Muh Taufik:** Writing – review & editing, Methodology. **Bagus P. Muljadi:** Writing – review & editing, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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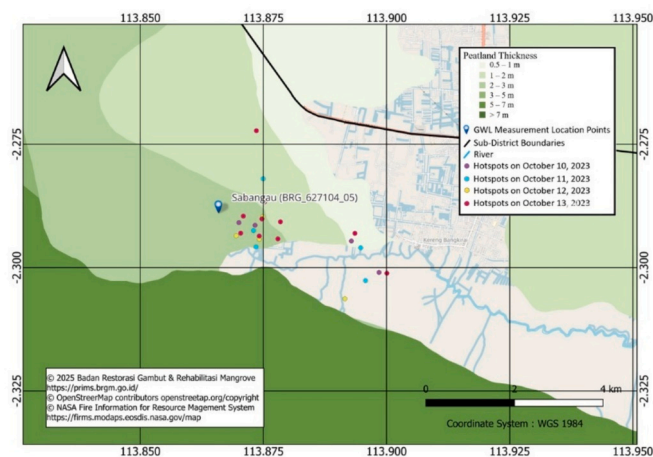


Fig. 9. The hotspots observed around the BRGM station in Sabangau, Central Kalimantan, from 10 October 2023 until 13 October 2023. The prediction from *peatfr* is consistent with the actual fire hotspots, suggesting the capability of this package as an early warning signal for tropical peatland fires.

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Data availability

Package *peatfr* is openly available on GitHub at <https://github.com/mellygsln/peatfr>. The time series data of water table, soil moisture, rainfall, and temperature employed in this study are openly available at <https://bit.ly/DatasetSabangau>.

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